

SHERKIN ISLAND, Ireland

WMO: 039510

Lat: 51.4767N

Lon: 9.4289W

Elev: 21

StdP: 101.07

Time Zone: 1.00 (IRL)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.2	1.3	-3.5	2.8	1.4	-2.1	3.2	2.6	19.2	8.6	17.4	8.9	5.3	50	0.737

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.2	19.6	16.6	18.3	15.8	17.6	15.4	17.2	18.6	16.6	17.8	16.1	17.1	5.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	11.9	17.6	16.2	11.5	17.1	15.7	11.1	16.7	48.5	18.6	46.8	17.9	45.2	17.1	19.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	14.3	12.8	DB	-1.1	20.9	2.5	2.0	-2.9	22.3	-4.3	23.5	-5.7	24.6	-7.5	26.0	(4)
(5)			WB	-1.7	18.0	2.3	1.1	-3.3	18.7	-4.7	19.4	-6.0	20.0	-7.7	20.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	7.7	7.5	7.7	9.4	11.5	13.8	15.2	15.0	14.1	12.2	9.3	8.3	(6)
(7)		DBStd	3.50	2.41	2.34	2.18	1.66	1.79	1.66	1.61	0.99	1.41	1.89	2.52	2.71	(7)
(8)		HDD10.0	361	75	73	73	29	5	0	0	0	0	4	40	60	(8)
(9)		HDD18.3	2680	331	304	329	269	212	137	97	105	127	190	269	309	(9)
(10)		CDD10.0	723	3	2	3	11	51	113	162	153	123	73	21	9	(10)
(11)		CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)
(12)		CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	6.3	7.2	7.2	6.5	6.1	5.5	5.4	5.0	5.5	6.0	6.5	7.5	7.9	(14)
(15)	Precipitation	PrecAvg	1280	150	112	87	79	79	84	94	88	95	131	136	146	(15)
(16)		PrecMax	1625	240	249	146	129	149	191	177	186	214	225	320	378	(16)
(17)		PrecMin	1019	58	27	24	23	4	15	41	24	28	30	38	34	(17)
(18)		PrecStd	166	50	63	37	34	36	36	36	49	43	55	57	68	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	11.9	13.4	14.8	19.2	21.3	22.0	18.7	18.5	16.7	14.5	12.8	(19)
(20)			MCWB	11.2	10.6	10.3	11.8	16.1	16.5	17.8	16.0	16.2	15.6	13.7	11.9	(20)
(21)		2%	DB	11.4	11.3	11.5	13.5	16.5	18.7	20.2	17.9	17.5	16.0	13.9	12.5	(21)
(22)			MCWB	10.6	10.3	9.9	11.2	13.8	15.5	17.2	15.7	15.6	15.1	13.1	11.6	(22)
(23)		5%	DB	11.0	10.9	11.0	12.4	15.1	17.6	19.0	17.3	16.8	15.3	13.2	12.0	(23)
(24)			MCWB	10.2	10.1	9.7	10.4	12.9	14.7	16.5	15.4	15.2	14.3	12.3	11.3	(24)
(25)		10%	DB	10.6	10.5	10.5	11.7	14.1	16.6	17.9	16.8	16.2	14.7	12.8	11.5	(25)
(26)			MCWB	9.8	9.7	9.3	10.0	12.3	14.2	15.8	15.1	14.8	13.6	11.7	10.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.2	11.2	12.9	15.7	17.0	18.6	17.4	17.1	16.6	14.1	12.5	(27)
(28)			MCDWB	11.8	11.4	11.7	14.3	18.2	20.3	20.7	17.9	17.6	16.6	14.3	12.7	(28)
(29)		2%	WB	11.1	10.8	10.7	11.7	14.2	16.0	17.7	16.7	16.3	15.6	13.4	12.0	(29)
(30)			MCDWB	11.3	11.1	11.1	12.6	15.9	18.1	19.4	17.3	16.8	15.8	13.7	12.3	(30)
(31)		5%	WB	10.6	10.4	10.3	11.2	13.4	15.3	17.0	16.2	15.9	14.7	12.7	11.6	(31)
(32)			MCDWB	10.9	10.8	10.9	11.9	14.7	17.0	18.5	16.8	16.4	15.1	13.2	11.9	(32)
(33)		10%	WB	10.1	10.0	9.8	10.7	12.7	14.7	16.3	15.7	15.4	14.0	12.1	11.0	(33)
(34)			MCDWB	10.6	10.5	10.5	11.4	13.8	16.2	17.7	16.4	16.0	14.6	12.8	11.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.5	3.5	4.1	4.4	4.3	4.3	4.2	3.9	3.7	3.6	3.6	3.5	(35)
(36)			MCDBR	2.9	2.8	4.0	4.8	5.7	5.9	5.6	4.4	4.2	3.6	3.2	3.1	(36)
(37)			MCWBR	3.4	3.0	3.6	3.5	3.7	3.7	3.6	3.1	3.3	3.2	3.5	3.6	(37)
(38)		5% WB	MCDBR	2.8	2.6	3.6	4.2	4.9	5.1	5.1	3.7	3.5	3.0	3.2	3.2	(38)
(39)			MCWBR	3.5	2.9	3.6	3.4	3.5	3.5	3.6	3.2	3.2	3.1	3.6	3.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.353	0.359	0.382	0.384	0.395	0.376	0.382	0.366	0.364	0.358	0.344	0.346	(40)	
(41)		taud	2.246	2.311	2.283	2.327	2.335	2.427	2.416	2.474	2.465	2.416	2.316	2.218	(41)	
(42)		Ebn at Noon	627	751	806	854	859	879	867	865	823	747	639	561	(42)	
(43)		Edh at Noon	71	87	107	114	119	109	109	98	89	77	65	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.79	1.49	2.64	4.27	5.17	5.67	5.15	4.39	3.17	1.82	1.00	0.61	(44)	
(45)		RadStd	0.08	0.12	0.27	0.29	0.31	0.45	0.42	0.25	0.25	0.17	0.07	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page